Course Name: Discrete Mathematics

Course Outcome

CO1: Understand the notion of mathematical thinking and proofs to solve the problem.
CO2: Apply the basics of discrete probability and number theory to solve the real world problem.
CO3: Analyze basic discrete structures and algorithms using effectively algebraic techniques.
CO4: Analyze mathematical concepts like sets, reasoning, relational algebra and graph theory to solve optimization problems.
CO5: Analyze the validity of an argument using logical notation.
CO6: Demonstrate the basic structures of proof techniques to write and evaluate the validity of arguments.
CO7: Understand the basic principles of sets, set equalities and operations in sets.
CO8: Apply counting principles to determine probabilities.

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Mid Term Examination, Odd Semester 2023-24
B. TECH(CS/AIML), II/III Year
BCSC 1010 DISCRETE MATHEMATICS

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	A computer company receives 350 applications from computer graduates for a job planning a line of new Web servers. Suppose that 220 of these applicants majored in computer science, 147 majored in business, and 51 majored both in computer science and in business. How many of these applicants majored neither in computer science nor in business?	3	CO4	E	P
2	How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen? How many must be selected to guarantee that at least three hearts are selected?	3	CO8	An	C
3	Show that the relation R on the set of all bit strings such that s R t if and only if s and t contain the same number of 1s is an equivalence relation.	3	CO7	E	P
4	(a) Suppose that there are eight runners in a race. The winner receives a gold medal, the second place finisher receives a silver medal, and the third-place finisher receives a bronze medal. How many different ways are there to award these medals, if all possible outcomes of the race can occur and there are no ties? (b) How many different 11 letter words are possible using the letters of the word COMBINATION such that 2 O's come together ?	3	CO8	An	C

5	Conjecture a formula for the sum of the first n positive odd integers. Then prove your conjecture using mathematical induction.	3	CO1	C	C

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Solve the given equation using generating function $a_n = 8a_{n-1} + 10^{n-1}$, $a_2 = 82$	5	CO8	E	P
7	Answer these questions for the poset ({3, 5, 9, 15, 24, 45}, |). a) Find the maximal elements. b) Find the minimal elements. c) Is there a greatest element? d) Is there a least element? e) Find all upper bounds of {3, 5}.	5	CO1	E	P
8	A popular puzzle of the late nineteenth century invented by the French mathematician Édouard Lucas, called the Tower of Hanoi, consists of three pegs mounted on a board together with disks of different sizes. Initially these disks are placed on the first peg in order of size, with the largest on the bottom. The rules of the puzzle allow disks to be moved one at a time from one peg to another as long as a disk is never placed on top of a smaller disk. The goal of the puzzle is to have all the disks on the second peg in order of size, with the largest on the bottom. Find the recurrence relation and solve it.	5	CO8	A	C